PROJECTION FORMULA FOR RF_!

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- j-shriek commutes with tensor (Cohomology on Sites, Lemma sites-cohomology-lemma-j-shriek-and-tensor, canonical tag 0GL1). cited at proof lines 3944-3947. j_!E tensor f-bar^{-1}K = j_!(E tensor f^{-1}K) for the open immersion.
- Projection formula for proper pushforward with torsion coefficients (Etale Cohomology, Lemma etale-cohomology-lemma-projection-formula-proper-mod-n, canonical tag 0F0G). cited at proof lines 3948-3950. The proper pushforward version of the projection formula applied to the compactification.

Source reference. Original lemma 0GL1: sources/stacks-chow/tags.txt, source lines 20706–20706. Cited at proof lines 3944-3947: j_!E tensor f-bar^{-1}K = j_!(E tensor f^{-1}K) for the open immersion.

Source reference. Original lemma 0F0G: sources/stacks-chow/tags.txt, source lines 18761–18761. Cited at proof lines 3948-3950: the proper pushforward version of the projection formula applied to the compactification.

Lemma 0.2. *Let $f : X \to Y$ be a separated finite type morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion ring. Let $E \in D(X_{étale}, \Lambda)$ and $K \in D(Y_{étale}, \Lambda)$. Then*

$$Rf_!E \otimes^{\mathbf{L}}_{\Lambda} K = Rf_!(E \otimes^{\mathbf{L}}_{\Lambda} f^{-1}K)$$

in $D(Y_{étale}, \Lambda)$.

Proof. Choose $j : X \to \overline{X}$ and $\overline{f} : \overline{X} \to Y$ as in the construction of $Rf_!$. We have $j_!E \otimes^{\mathbf{L}}_{\Lambda} \overline{f}^{-1}K = j_!(E \otimes^{\mathbf{L}}_{\Lambda} f^{-1}K)$ by Cohomology on Sites, Lemma 0GL1. Then we get the result by applying Étale Cohomology, Lemma 0F0G and using that $f^{-1} = j^{-1} \circ \overline{f}^{-1}$ and $Rf_! = R\overline{f}_* j_!$. □

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